Post-Quantum Mathematics

Part 2 - Standards Tracks & Candidates: Non-NIST National & Regional Standards Tracks

Compiled by the Spaced Research Ledger

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NIST is not the only body choosing post-quantum algorithms. South Korea ran its own competition and picked its own winners. China is running a second one and has said openly that it intends to shape international norms rather than adopt American standards. Germany and France endorse algorithms NIST did not select, and require them in combination rather than alone.

This matters for anyone building something that has to work in more than one country. The algorithms are converging, but the requirements around them are not. A product certified in France must use a hybrid construction. A product sold to the German government faces a 2030 deadline for its most sensitive uses. A product intended for the Korean public sector will need algorithms that appear in no NIST document.

One term recurs. A hybrid deployment runs a post-quantum algorithm alongside a classical one and combines the results, so that breaking the system requires breaking both. Several of the bodies below require this. NIST permits it but does not mandate it, and that is the single largest divergence in this part.

Section 1 covers South Korea, Section 2 China, Section 3 Germany, Section 4 France and the wider EU, Section 5 Japan, and Section 6 the European standards institute ETSI.

1. South Korea

South Korea ran a full national competition and selected four winners of its own. It is now moving them into silicon, which is further than any other non-NIST track in this part has gone.

Recent Developments

ICTK was selected in June 2026 to lead a national research project worth 7 billion won, running from April 2026 to December 2029 [12]. The work implements ML-KEM and ML-DSA alongside the Korean schemes SMAUG-T and HAETAE in a security chip, with Korean cryptographic module certification.

That combination is the notable part. The chip carries both the NIST algorithms and the national ones, which is what practical divergence looks like in hardware.

Official specification documents and reference code for HAETAE and SMAUG-T were published in January 2026 as part of finalizing the Korean selections [11].

Current State

The competition is run by South Korea’s National Intelligence Service with the National Security Research Institute [1]. It has run since 2021, as part of the country’s 2023 post-quantum master plan.

South Korea announced its final four winners in January 2025 [1]. For digital signatures it chose HAETAE and AIMer. For public-key encryption and key encapsulation it chose SMAUG-T and NTRU+. AIMer was developed jointly by Samsung SDS and the Korea Advanced Institute of Science and Technology, and won in the signature category [2].

The algorithms are set for use in Korean government systems, with a transition roadmap targeting 2035, which the source describes as aligned with American and European timelines [1].

2. China

China ran an early competition, set it aside, and started again with a new process open to the world. Its stated aim is to shape international norms rather than adopt NIST's choices.

Recent Developments

China's Institute of Commercial Cryptography Standards opened a submission window for public-key and hash algorithms running from October 2025 to June 2026 [13].

Current State

That window belongs to a call launched in February 2025, open internationally rather than restricted to Chinese developers [4]. The restriction is what distinguishes it from the earlier effort.

As of March 2026, China expects its own post-quantum standards within three years, implying roughly 2028 or 2029 [5]. No algorithm had been selected from the 2025 call as of this research. China is reported to be prioritizing the finance and energy sectors, and to be pursuing distinct technical approaches including unstructured lattice algorithms [5]. Unstructured lattices are the same conservative choice Germany makes in Section 3, for the same reason: less algebraic structure means less for a future attack to exploit.

One correction belongs in the record. An earlier competition did run from 2018 to 2019, restricted to Chinese developers, with 36 submissions [3]. There is no evidence that its candidates represent China's current national direction, and the country's approach has since moved to the separate process described above. The earlier competition is historical background only.

3. Germany

Germany's federal security office publishes its own recommendations, and has done so since well before NIST finished. It endorses an algorithm NIST rejected, and it has set harder deadlines than the American ones.

Current State

The office published version 2026-01 of its technical guideline on cryptographic mechanisms, with post-quantum cryptography as a current focus [6].

The endorsements diverge from NIST deliberately. The guideline backs ML-KEM and FrodoKEM for key exchange, and the stateful hash-based schemes XMSS and LMS for signatures [6]. Its recommendations for FrodoKEM and Classic McEliece date back to 2020, predating NIST's own selections. Both of those schemes are covered in Part 3, and neither was chosen by NIST.

The deadlines are firm. Sole use of classical key agreement is permitted only until 31 December 2031, and applications needing very high protection must complete migration by 31 December 2030 [6]. The office continues to recommend hybrid deployment throughout the transition.

4. France and the European Union

France has attached post-quantum requirements to product certification, which converts guidance into a market condition. Its agency also reserves the right to certify algorithms NIST did not pick.

Recent Developments

France's national cybersecurity agency announced in June 2026 that it will stop certifying security products lacking quantum-resistant encryption, starting in 2027 [17].

Current State

Certification is not optional for the buyers who matter. It is a prerequisite for use across French government agencies and critical-infrastructure operators, and the agency separately expects businesses to be purchasing only quantum-safe products by 2030 [8].

The technical requirements are stricter than NIST's. The agency requires hybrid implementations generally, with limited exceptions for the hash-based signature schemes XMSS, LMS and SLH-DSA [7]. For key exchange it recommends ML-KEM combined with FrodoKEM, and for signatures it requires ML-DSA paired with a classical scheme. It has explicitly reserved the right to certify an algorithm set that does not match NIST's, naming FrodoKEM as an acceptable alternative [8].

The wider European timeline sits above all of this. France, Germany and the Netherlands jointly led the EU's coordinated implementation roadmap, published in June 2025 [7]. It sets three milestones. National transition plans are due by the end of 2026, high-risk systems migrated by the end of 2030, and full migration by the end of 2035.

5. Japan

Japan added ML-KEM to its recommended ciphers list in 2026, and is pursuing post-quantum cryptography alongside quantum key distribution rather than instead of it.

Recent Developments

ML-KEM, in its 768 and 1024 parameter sets, was added to Japan's recommended ciphers list in the March 2026 update [16].

Current State

That listing followed an external evaluation. PQShield announced in April 2026 that it had delivered an ML-KEM evaluation to Japan's cryptographic standardization body [9]. The evaluation supported the algorithm's inclusion, and is described as removing a major barrier to adoption. The listing is described as the first clear official signal of Japanese government acceleration, ahead of a formal national roadmap expected by May 2027 [9].

Japan is pursuing a dual-track strategy, running post-quantum cryptography alongside quantum key distribution, with a government-funded project from 2024 to 2026 advancing implementation across the technology supply chain [10]. No other country in this part is pursuing both tracks together in this way.

6. ETSI

The European standards institute works on the layer NIST does not cover: how to implement these algorithms without introducing new weaknesses.

Recent Developments

ETSI published a technical report in May 2026 on secure implementation guidance for key encapsulation mechanisms and digital signature schemes [15].

A new ETSI committee on quantum technologies held its first meeting in December 2025, with elected leadership [14]. Its initial work items cover mapping the quantum ecosystem and guidelines for quantum random number generators.

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